

Toolmakers: They make dies and molding tools which are used to punch and form these components and produce plastic components such as car bumpers.

Fitters: They assemble components into larger product.

Maintenance fitters: They repair machinery.

Welders: They do specialized joining, fabricating, and repair work.

Electricians: They wire and install electrical equipment.

Operators: They require fewer skills. Many operator jobs consist mainly of minding a machine, especially now that more and more processes are automated. However, some operators may have to check components produced by their machines to ensure they are accurate. They may require training in the use of instruments such as micrometers, verniers, or simple 'go/no go' gauges.

Who is employed to do the following engineering jobs? Read the given text to decide your answer. Write your answer in the blanks given.

Text A

1. Seeing that the correct test equipment is available on a production line:

2. Finding a cheaper way of manufacturing a crankshaft:

Text B

3. Testing completed motors from a production line:

4. Finding out why a new electronics assembly does not work:

5. Repairing heating systems installed by their company:

6. Seeing that a new product is safe to use:

Text C

7. Producing a mould for a car body part:

8. Starting a turbine in a power station:

Refer to all texts (A, B, C) and identify the engineering jobs of the following workers from their statements.

Worker 1: We perform standard chemical and physical tests on samples, usually as a result of a complaint from inspectors on the production line. We are an important part of production. We have the authority to stop the line if we find something seriously wrong. It is interesting work, and we're able to move around from test to test and chat. Sometimes, admittedly, the work gets a bit repetitive.

Answer: _____

Worker 2: All machinists can be difficult. The older blokes especially don't like me telling them their work isn't good enough and instructing them to do it again. One or two of them seem to think the inspector is always out to get them. I constantly have to calm things down.

Answer: _____

Worker 3: My company makes desalination equipment. It takes the salt out of sea water so that it can be used for drinking and irrigation. A lot of our customers are in the Middle East. I have to go there whenever new equipment is being set up to make sure it's properly installed and everything is running ok.

Answer: _____

Worker 4: I enjoy my job. I really like doing the same thing every day—exactly the same job. You know what to look for and how things should be. You know how the machine—or the machines—run, when a machine is working properly and when there is something wrong with it. I really like the routine. I don't have dreams of becoming a supervisor or anything like that. I'm just content running my machines.

Answer: _____

Answer the following questions by writing Worker 1, or Worker 2, or Worker 3, or Worker 4 in the blanks.

1. Which worker suggests that he/she does not enjoy his/her job as much as the other workers? _____
2. Whose job involves travelling? _____
3. Who hints that he/she enjoys his/her job despite its being at times monotonous? _____

Part Two: Vocabulary from Text (10%)

A. Complete the sentence in the following paragraph using the most appropriate word from the list below. (5%)

mechanisms, manufacturing, downloaded, automated, invented, innovations, technological, gadgets, surf, browsing, technologically, mass-produce, mechanically

Technology has changed rapidly and every ¹ _____ advancement has led to better service for mankind than preceding ones. Quite a large number of technicians have ² _____ many materials in due processes. These ³ _____ have made the world a great and convenient place to live in. For example, agricultural processes that once required large number of workers have now been ⁴ _____ which means cost efficiency for farmers. The ⁵ _____ industry has also been greatly modified as many companies that can ⁶ _____ products in a short period of time have been established across the world. Similarly, regarding entertainment and information, a couple of decades ago, people ^{used} to listen to radios. Afterwards, albums and cassettes came into the spotlight. At present these ⁷ _____ are replaced by CDs and the internet. Thus, music ⁸ _____ from internet can be played using MP3. Besides, cell phones can be used to ⁹ _____ the internet. One may bet there will be other ¹⁰ _____ in five years time.

B. Complete the following sentences by using the most appropriate item from the list below. (5%)

generate, affect, durability, extrusion, sketch, hack, effect, career, technician, conducting, download, sketch, rivals, installed, credit card, criminal, low-tech, protect, incapacitate, commit

1. Although Mr. Milki has _____ security camera, his house could not be spared from robbery.
2. These days the number of people trying to _____ bank accounts is on the rise.
3. There are many technology graduates in our country that are engaged in producing _____ products.
4. Global warming has severe _____ on weather condition.
5. Policemen usually _____ dangerous offenders using different kinds of technological devices.

6. Using _____ will undoubtedly reduce the risk of theft because one can make large transactions without necessarily carrying large sum of cash.
7. Though utensils made of plastic are cheaper their _____ is the major concern for users.
8. Although technology is becoming a useful tool in fighting illegal acts, the offenders are always designing mechanisms that help them to _____ crime.
9. It is believed that the Grand Ethiopian Renaissance dam will _____ more electric power than the rest of Ethiopia's power stations.
10. Industrial designers usually produce the _____ of different products from things around us.

Part Three Grammar (10%)

A. Use present active or present passive form of the verbs in the parenthesis.(3.5Pts.)

Computational Fluid Dynamics is computer software used to assist in aerodynamic design. It
 1 _____ (model) the flow of air over surface, such as car body work or the fuselage and
 wings of aircraft. Virtual testing with CFD software 2 _____ (do) in the early stages of the
 design process. Wind tunnels equipped with rolling roads 3 _____ (allow) reduced scale
 models of vehicles or full-size vehicle to be tested. Air 4 _____ (blow) through the tunnel by
 powerful fans to create airflow of different velocities which simulate the vehicle travelling at
 different speed. The airflow over the surface of the vehicle 5 _____ (highlight), so that it
 6 _____ (can analyze). A rolling road is effectively a conveyor belt which moves
 beneath the stationary vehicle at the same speed as the air flow, making the wheels turn. This
 7 _____ (allow) engineers to analyze the effects of the spinning wheels on the air flow

B. Using the information in the following table, write five comparative degree sentences which compare the two types of smart phone.(2Pts)

	Galaxy S4 Samsung	I phone 5 S Apple
Size	5'' (inch)	4'' (inch)
Battery capacity	2,600 mA·h	1,570 mA·h
Camera resolution	12.8 MP	8 MP
Weight	130 g	112 g
RAM	1 GB	2GB
Price	\$399	\$431

1. _____
2. _____
3. _____
4. _____

Change the following statements into Yes/No questions. (2Pts)

1. The machine needs regular lubrication.

2. Kevlar is wear resistant.

3. METEC built the prototype of drones.

4. Our country will achieve the millennium goal.

Ask information questions to get the answers.(2Pts)

1. Where _____
She works in METEC.
2. What _____
She is a designer.
3. What _____
She designs army vehicle parts.
4. Who _____
She works with three technicians.

Part Four Writing (5%)

A/ The following statements describe the stages in the process of operating a washing machine. But they are in a jumbled order. Read the sentences and rearrange them in the correct order by writing the letters of the sentences on the given space. The first one is given you as example. (5%)

First, the drum is filled with water.

- a. Next the soap is added and the drum is rotated slowly.
- b. Now the drum is rotated much faster and the water pumped out.
- c. Finally, the clean clothes are removed.
- d. Then the water is heated to the right temperature.
- e. After that the dirt water is pumped out and clean water is added.

The correct order is 1 _____ 2. _____ 3. _____ 4. _____ 5. _____